

Examen Parcial 12/05/2023

1. Si $\nabla \cdot \mathbf{v} = 0$, $\nabla \times \mathbf{v} = \dot{\mathbf{w}}$ y $\nabla \times \mathbf{w} = -\dot{\mathbf{v}}$, donde $\dot{\mathbf{v}} = \frac{d\mathbf{v}}{dt}$ y $\dot{\mathbf{w}} = \frac{d\mathbf{w}}{dt}$, demostrar usando notación indicial que

$$\nabla^2 \mathbf{v} = \ddot{\mathbf{v}}$$

donde $\ddot{\mathbf{v}} = \frac{d^2 \mathbf{v}}{dt^2}$ y donde la componente j -ésima del vector $\nabla^2 \mathbf{v}$ es $[\nabla^2 \mathbf{v}]_j = \frac{\partial^2 v_j}{\partial x_i \partial x_i}$.

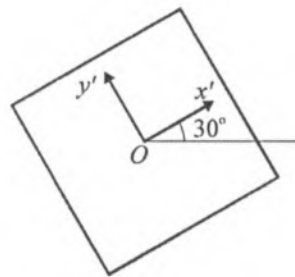
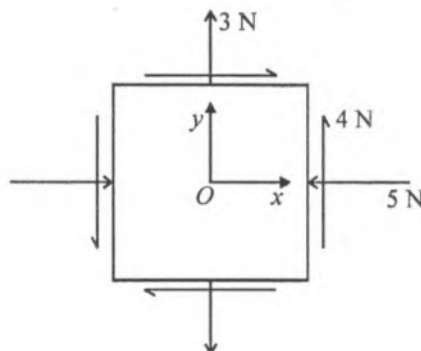
2. Sea $O-x_1x_2x_3$ un sistema de coordenadas Cartesianas rectangulares, cuya base es $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. Se define un nuevo sistema de coordenadas Cartesianas rectangulares $O'-x_1'x_2'x_3'$ de base $\{\mathbf{e}_1', \mathbf{e}_2', \mathbf{e}_3'\}$, con $O'(4, -1, -2)$, $\mathbf{e}_1' = (2\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3)/3$ y $\mathbf{e}_2' = (\mathbf{e}_1 - \mathbf{e}_2)/\sqrt{2}$.
- Expresar $\mathbf{e}_3'$ en términos de $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.
 - Dado $\mathbf{t} = 10\mathbf{e}_1 + 10\mathbf{e}_2 - 20\mathbf{e}_3$, expresar $\mathbf{t}$ en la base $\{\mathbf{e}_1', \mathbf{e}_2', \mathbf{e}_3'\}$.
 - Expresar las coordenadas originales x_1, x_2, x_3 en términos de las nuevas x_1', x_2', x_3' .

3. Respecto a los ejes coordenados $\mathbf{e}_1, \mathbf{e}_2$ y $\mathbf{e}_3$, el tensor de tensión en un punto tiene los componentes:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

- Determine la componente normal σ_n y la tangencial τ del vector tracción $\mathbf{T}$ en el plano de normal $\mathbf{v} = (\mathbf{e}_1 + 3\mathbf{e}_2 - \mathbf{e}_3)/\sqrt{11}$.
- Determine las componentes del vector unitario $\mathbf{e}_\tau$ tal que $\mathbf{T} = \sigma_n \mathbf{v} + \tau \mathbf{e}_\tau$.

4. Sea el estado plano de tensiones de la figura de la izquierda:



- Calcular las tensiones y direcciones principales.
- Calcular dirección y magnitud de la máxima tensión de corte.
- Calcular el estado de tensiones correspondiente a la figura de la derecha.

$$\nabla \cdot \underline{v} = 0$$

$$\nabla \times \underline{v} = \underline{w}$$

$$\nabla \times \underline{w} = -\underline{v}$$

$$\nabla^2 \underline{v} = \underline{v} \quad ?$$

$$\underline{v} = \frac{d\underline{v}}{dt}$$

$$\underline{w} = \frac{d\underline{w}}{dt}$$

$$v_i = \frac{d}{dt} \frac{dv_i}{dt} = \frac{d}{dt} (-\epsilon_{ijk} w_{k,j}) =$$

$$= -\epsilon_{ijk} \frac{\partial}{\partial x_j} \frac{d}{dt} w_k = -\epsilon_{ijk} \frac{\partial}{\partial x_j} \epsilon_{klm} \frac{\partial v_m}{\partial x_l} =$$

$$= -\epsilon_{kij} \epsilon_{klm} \frac{\partial^2}{\partial x_l \partial x_j} v_m =$$

$$= -(\delta_{il} \delta_{jm} - \delta_{jl} \delta_{im}) \frac{\partial^2}{\partial x_l \partial x_j} v_m =$$

$$= -\frac{\partial^2}{\partial x_i \partial x_m} v_m + \frac{\partial^2}{\partial x_j \partial x_j} v_i =$$

$$= -\frac{\partial}{\partial x_i} \underbrace{\frac{\partial v_m}{\partial x_m}}_0 + \frac{\partial^2}{\partial x_j \partial x_j} v_i = (\nabla^2 \underline{v})_i$$

$$2) \quad O = x_1 - x_2 - x_3$$

$$\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$$

$$O' = x'_1 - x'_2 - x'_3 \quad \{\underline{e}'_1, \underline{e}'_2, \underline{e}'_3\}$$

$$O' (1, -1, -2)$$

$$\underline{e}'_1 = (2\underline{e}_1 + 2\underline{e}_2 + \underline{e}_3)/3$$

$$\underline{e}'_2 = (\underline{e}_1 - \underline{e}_2)/\sqrt{2}$$

$$a) \underline{e}'_3 = ?$$

$$\underline{e}'_3 = \underline{e}'_1 \times \underline{e}'_2 =$$

$$= (2\underline{e}_1 + 2\underline{e}_2 + \underline{e}_3) \times (\underline{e}_1 - \underline{e}_2) / 3\sqrt{2} =$$

$$= \frac{1}{3\sqrt{2}} (-2\underline{e}_1 \times \underline{e}_2 + 2\underline{e}_2 \times \underline{e}_1 + \underline{e}_3 \times \underline{e}_1 - \underline{e}_3 \times \underline{e}_2)$$

$$= \frac{1}{3\sqrt{2}} (-2\underline{e}_3 - 2\underline{e}_3 + \underline{e}_2 + \underline{e}_1) =$$

$$\underline{e}'_3 = \frac{1}{3\sqrt{2}} (\underline{e}_1 + \underline{e}_2 - 4\underline{e}_3)$$

$$b) \underline{t} = 20\underline{e}_1 + 20\underline{e}_2 - 20\underline{e}_3$$

$$[\underline{e}'_1 \quad \underline{e}'_2 \quad \underline{e}'_3] = [\underline{e}_1 \quad \underline{e}_2 \quad \underline{e}_3] \times \begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/3\sqrt{2} \\ 2/3 & -1/\sqrt{2} & 1/3\sqrt{2} \\ 1/3 & 0 & -1/3\sqrt{2} \end{bmatrix}$$

$$\underline{\underline{A}}^{-1} = \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -1/3\sqrt{2} \end{bmatrix} = A^T$$

$\underline{\underline{A}}$

(A orthogonal)

$$[\underline{e}_1 \underline{e}_2 \underline{e}_3] = [\underline{e}'_1 \underline{e}'_2 \underline{e}'_3] \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -1/3\sqrt{2} \end{bmatrix}$$

$$\underline{t} = [\underline{e}_1 \underline{e}_2 \underline{e}_3] \begin{pmatrix} 10 \\ 10 \\ -20 \end{pmatrix} = [\underline{e}'_1 \underline{e}'_2 \underline{e}'_3] \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -1/3\sqrt{2} \end{bmatrix} \begin{pmatrix} 10 \\ 10 \\ -20 \end{pmatrix}$$

$$\underline{t} = [\underline{e}'_1 \underline{e}'_2 \underline{e}'_3] \begin{pmatrix} 20/3 \\ 0 \\ +100/3\sqrt{2} \end{pmatrix}$$

c)

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/3\sqrt{2} \\ 2/3 & -1/\sqrt{2} & 1/3\sqrt{2} \\ 1/3 & 0 & -1/3\sqrt{2} \end{bmatrix} \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix}$$

$$3] \quad \underline{\underline{G}} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

a) $\underline{\underline{G}}_n$ y $\underline{\underline{r}}$ en plano de normal $\underline{\underline{r}} = (e_1 + 3e_2 - e_3)/\sqrt{11}$

$$\underline{\underline{T}} = \underline{\underline{G}} \underline{\underline{r}} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} \frac{1}{\sqrt{11}} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix}$$

$$\underline{\underline{G}}_n = \underline{\underline{T}} \cdot \underline{\underline{r}} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} \frac{1}{\sqrt{11}} = \frac{1}{11} (2 + 27 + 4) =$$

$$\underline{\underline{r}} = \underline{\underline{T}} - \underline{\underline{G}}_n \underline{\underline{r}} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix} - \frac{3}{\sqrt{11}} \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} =$$

$$= \frac{1}{\sqrt{11}} \begin{pmatrix} 2-3 \\ 9-9 \\ -4+3 \end{pmatrix} = \frac{1}{\sqrt{11}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

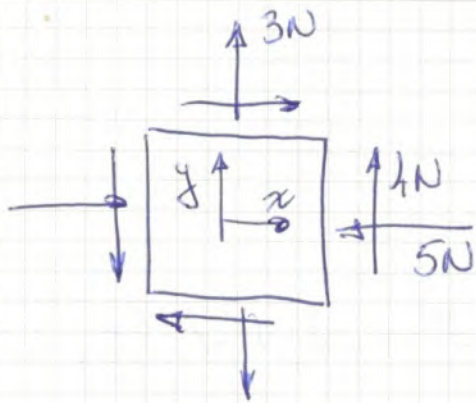
$$\underline{\underline{r}} = |\underline{\underline{r}}| = \sqrt{\frac{2}{11}}$$

b)

$$\underline{\underline{e}}_r = \frac{\underline{\underline{r}}}{\underline{\underline{r}}} = \frac{1}{\sqrt{11}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \frac{\sqrt{11}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

4

a)



$$\underline{\underline{\sigma}} = \begin{bmatrix} -5 & 4 \\ 4 & 3 \end{bmatrix}$$

$$(\underline{\underline{\sigma}} - \sigma \underline{\underline{I}}) \underline{\underline{v}} = 0$$

$$\det \begin{pmatrix} -5 - \sigma & 4 \\ 4 & 3 - \sigma \end{pmatrix} = 0$$

$$(-5 - \sigma)(3 - \sigma) - 16 = 0$$

$$-15 + 5\sigma - 3\sigma + \sigma^2 - 16 = 0$$

$$\sigma^2 + 2\sigma - 31 = 0$$

$$\sigma_{1,2} = \frac{-2 \pm \sqrt{4 + 4 \times 31}}{2} = \frac{-2 \pm 8\sqrt{2}}{2} = -1 \pm 4\sqrt{2}$$

$$\sigma_1 = \sigma_{\max} = 4\sqrt{2} - 1 \approx 4,656 \dots$$

$$\sigma_2 = \sigma_{\min} = -4\sqrt{2} - 1 \approx -6,656 \dots$$

ooo